

Mathematical Analysis of an SIR-Type Model of Misinformation Spread with Educational-Campaign Control

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Abstract

This note collects the mathematical results for the SCM (Susceptible–Critically-literate–Misinformed) model of misinformation spread under a time-dependent educational-campaign control $E(t)$: positivity and boundedness of solutions, the misinformation-free equilibrium, the control reproduction number via the next-generation matrix method, local stability of that equilibrium, and the existence and characterization of an optimal campaign via Pontryagin’s Maximum Principle. Proofs are given in full.

1 The SCM Model

Because misinformation cascades on breaking news typically unfold over a rapid timescale (hours to days), we assume demographic turnover is negligible over the simulation window and take a closed population with no vital dynamics (recruitment rate $\Lambda = 0$ and natural exit rate $\mu = 0$). The resulting system of ordinary differential equations is

$$\frac{dS}{dt} = -\alpha_1 ES + \psi C - \beta SM + \theta M, \quad (1)$$

$$\frac{dC}{dt} = \alpha_1 ES - \psi C + \gamma CM + \alpha_2 EM, \quad (2)$$

$$\frac{dM}{dt} = \beta SM - \gamma CM - \theta M - \alpha_2 EM - dM. \quad (3)$$

All state variables, the control, and the parameters (Table 1) are assumed non-negative.

2 Well-Posedness: Positivity and Boundedness

We first establish that system (1)–(3) is mathematically well-posed and biologically meaningful: solutions remain non-negative and bounded for all $t \geq 0$.

Positivity of solutions

Proposition 1 (Positivity). *Given non-negative initial conditions $S(0), C(0), M(0) \geq 0$ and non-negative parameters, the solution $(S(t), C(t), M(t))$ of system (1)–(3) satisfies $S(t), C(t), M(t) \geq 0$ for all $t \geq 0$.*

Table 1: State variables, control, and parameters of the SCM model.

Symbol	Description	Units
$S(t)$	Susceptible population proportion	dimensionless
$C(t)$	Critically literate population proportion	dimensionless
$M(t)$	Misinformed population proportion	dimensionless
$E(t)$	Educational-campaign intensity (control)	dimensionless, $[0, E_{\max}]$
β	Transmission rate, $S \rightarrow M$ via contact with M	hour ⁻¹
γ	Debunking rate, $M \rightarrow C$ via contact with C	hour ⁻¹
ψ	Natural forgetting rate, $C \rightarrow S$	hour ⁻¹
θ	Natural forgetting rate, $M \rightarrow S$	hour ⁻¹
α_1	Campaign effectiveness on the susceptible ($S \rightarrow C$)	hour ⁻¹
α_2	Campaign effectiveness on the misinformed ($M \rightarrow C$)	hour ⁻¹
d	Disengagement (“too far gone”) exit rate from M	hour ⁻¹

Proof. The right-hand side of (1)–(3) is continuous and locally Lipschitz in (S, C, M) on $\mathbb{R}^3$, so a solution exists and is unique on some maximal interval $[0, t_{\max})$. We examine the derivative of each compartment on the boundary where its value is zero. If $S = 0$,

$$\left. \frac{dS}{dt} \right|_{S=0} = \psi C + \theta M \geq 0,$$

since $\psi, \theta, C, M \geq 0$. If $C = 0$,

$$\left. \frac{dC}{dt} \right|_{C=0} = \alpha_1 E S + \alpha_2 E M \geq 0,$$

since $\alpha_1, \alpha_2, E, S, M \geq 0$. If $M = 0$,

$$\left. \frac{dM}{dt} \right|_{M=0} = 0.$$

In each case, the vector field points into (or along the boundary of) the non-negative orthant whenever a compartment reaches zero, so no compartment can cross into negative values. Hence $S(t), C(t), M(t) \geq 0$ for all $t \in [0, t_{\max})$, and since solutions cannot blow up in finite time on a bounded region (Proposition 2), $t_{\max} = \infty$. $\square$

Boundedness of the population

Proposition 2 (Boundedness). *Let $N(t) = S(t) + C(t) + M(t)$. Then $N(t) \leq N(0)$ for all $t \geq 0$, so all solutions of (1)–(3) with non-negative initial conditions remain in the biologically feasible region $\Gamma = \{(S, C, M) \in \mathbb{R}_{\geq 0}^3 : S + C + M \leq N(0)\}$, which is positively invariant.*

Proof. Summing (1)–(3),

$$\begin{aligned} \frac{dN}{dt} &= \frac{dS}{dt} + \frac{dC}{dt} + \frac{dM}{dt} \\ &= (-\alpha_1 E S + \psi C - \beta S M + \theta M) + (\alpha_1 E S - \psi C + \gamma C M + \alpha_2 E M) \\ &\quad + (\beta S M - \gamma C M - \theta M - \alpha_2 E M - d M) \\ &= -d M. \end{aligned}$$

Because $d \geq 0$ and $M(t) \geq 0$ by Proposition 1, we have $dN/dt = -dM \leq 0$, so $N(t)$ is monotonically non-increasing. Hence $N(t) \leq N(0)$ for all $t \geq 0$, and combined with Proposition 1, every solution starting in Γ remains in Γ . $\square$

3 The Misinformation-Free Equilibrium and Reproduction Number

The misinformation-free equilibrium

At the misinformation-free equilibrium (MFE), $M = 0$ and the educational effort E is held constant. Let $N = S + C$ denote the (conserved, while $M = 0$) total of the remaining two compartments. Setting the right-hand sides of (1)–(2) to zero with $M = 0$,

$$-\alpha_1 ES + \psi C = 0, \quad (4)$$

$$\alpha_1 ES - \psi C = 0. \quad (5)$$

From (4), $C = (\alpha_1 E / \psi) S$. Substituting into $N = S + C$ and solving,

$$S^* = \frac{N\psi}{\psi + \alpha_1 E}, \quad C^* = \frac{N\alpha_1 E}{\psi + \alpha_1 E}.$$

Thus the misinformation-free equilibrium is

$$E_0 = \left(\frac{N\psi}{\psi + \alpha_1 E}, \frac{N\alpha_1 E}{\psi + \alpha_1 E}, 0 \right).$$

The control reproduction number

Because S^* and C^* depend on the constant campaign level E , the threshold quantity derived from them is a *control reproduction number* $R_c(E)$, with the basic reproduction number recovered as the no-campaign case $R_0 = R_c(0)$ (van den Driessche and Watmough, 2002). We use the next-generation matrix (NGM) method, decomposing the misinformed-compartment equation $dM/dt = F - V$ into new-misinformation and outflow terms:

$$F = \beta SM, \quad V = \gamma CM + \theta M + \alpha_2 EM + dM.$$

Taking partial derivatives with respect to M and evaluating at E_0 ,

$$F = \frac{\partial F}{\partial M} \Big|_{E_0} = \beta S^*, \quad V = \frac{\partial V}{\partial M} \Big|_{E_0} = \gamma C^* + \theta + \alpha_2 E + d.$$

Since F and V are 1×1 , the control reproduction number is the spectral radius of FV^{-1} :

$$R_c(E) = \frac{\beta S^*}{\gamma C^* + \theta + \alpha_2 E + d}. \quad (6)$$

At the model's calibrated parameters with no campaign ($E = 0$, so $S^* = N$ and $C^* = 0$), the basic reproduction number is $R_0 = R_c(0) = \beta/(\theta + d) \approx 1.56$, so the misinformation-free equilibrium is unstable and the rumour spreads; with $\alpha_1 = \alpha_2 = 0.1$, a sustained campaign of intensity $E \gtrsim 0.07$ already brings $R_c(E)$ below the critical threshold of 1.

4 Local Stability of the Misinformation-Free Equilibrium

Proposition 3 (Local stability). *Fix a constant campaign level E and let $N = S(0) + C(0)$. If $R_c(E) < 1$, then every trajectory of (1)–(3) with $M(0)$ sufficiently small and $S(0) + C(0) = N$ converges to the misinformation-free equilibrium $E_0 = (S^*, C^*, 0)$ as $t \rightarrow \infty$. If $R_c(E) > 1$, E_0 is unstable.*

To establish this, we linearize about E_0 . Writing $f_1 = dS/dt$, $f_2 = dC/dt$, $f_3 = dM/dt$, the Jacobian is

$$J = \begin{pmatrix} -(\alpha_1 E + \beta M) & \psi & -\beta S + \theta \\ \alpha_1 E & -(\psi - \gamma M) & \gamma C + \alpha_2 E \\ \beta M & -\gamma M & \beta S - \gamma C - (\theta + \alpha_2 E + d) \end{pmatrix}.$$

Evaluating at $E_0 = (S^*, C^*, 0)$, the matrix simplifies to block upper-triangular form:

$$J(E_0) = \begin{pmatrix} -\alpha_1 E & \psi & -\beta S^* + \theta \\ \alpha_1 E & -\psi & \gamma C^* + \alpha_2 E \\ 0 & 0 & \beta S^* - \gamma C^* - \theta - \alpha_2 E - d \end{pmatrix}.$$

The eigenvalues are determined by the 1×1 block $\lambda_M = \beta S^* - (\gamma C^* + \theta + \alpha_2 E + d) = F - V$, and the 2×2 block

$$J_{11} = \begin{pmatrix} -\alpha_1 E & \psi \\ \alpha_1 E & -\psi \end{pmatrix}.$$

Remark 1 (A structural degeneracy, and how we resolve it). Since $\text{tr}(J_{11}) = -(\alpha_1 E + \psi) < 0$ while $\det(J_{11}) = 0$ identically, J_{11} has eigenvalues $\lambda_1 = 0$ and $\lambda_2 = -(\alpha_1 E + \psi) < 0$. This is *not* an algebra error: with $\Lambda = \mu = 0$, setting $M \equiv 0$ in (1)–(2) conserves $S + C$, so E_0 is not isolated but lies on a line of equilibria $\mathcal{L}_E = \{(S, C, 0) : S + C = N\}$ indexed by the conserved total N . The zero eigenvalue makes E_0 non-hyperbolic, so Hartman–Grobman does not apply and “asymptotically stable” must be read relative to $\mathcal{L}_E$ rather than as an isolated point — a degeneracy standard in demography-free compartmental models (cf. the disease-free line in an SIS model without vital dynamics). The three eigenvalues have clear roles: $\lambda_1 = 0$ is the neutral direction selecting *which* line $\mathcal{L}_E$ a trajectory lands on (set by the population lost through $-dM$ during any excursion with $M > 0$); $\lambda_2 < 0$ governs relaxation *within* a line toward (S^*, C^*) ; and λ_M governs growth or decay of a small perturbation $M(0) > 0$ off the line.

Proof of Proposition 3. By Remark 1, $\lambda_2 < 0$ unconditionally, so it remains to sign λ_M . By definition, $R_c(E) = F/V$ with $F, V > 0$, so $R_c(E) < 1 \iff F < V \iff \lambda_M = F - V < 0$. Thus for $R_c(E) < 1$, both non-neutral eigenvalues at E_0 are strictly negative: a small perturbation with $M(0) > 0$ decays monotonically ($\lambda_M < 0$), and any transient in the (S, C) -directions within the corresponding line $\mathcal{L}_E$ relaxes to (S^*, C^*) ($\lambda_2 < 0$). Hence trajectories starting sufficiently close to $\mathcal{L}_E$ converge to the equilibrium point on $\mathcal{L}_E$ determined by their own conserved population level, establishing local asymptotic stability of E_0 relative to $\mathcal{L}_E$. Conversely, if $R_c(E) > 1$ then $\lambda_M > 0$ and small perturbations with $M(0) > 0$ grow, so E_0 is unstable. $\square$

Remark 2. Biologically, Proposition 3 says that a rumor introduced into a population with $R_c(E) < 1$ dies out on its own, while one introduced into a population with $R_c(E) > 1$ initially grows — the threshold behavior separated by $R_c(E) = 1$.

5 The Optimal Control Problem

Objective functional

We seek a time-dependent educational-campaign strategy $E(t)$ that minimizes the number of misinformed individuals over a fixed horizon $[0, t_f]$ while keeping the cost of the campaign low. We define the objective functional

$$J(E) = \int_0^{t_f} (M(t) + f(E(t))) dt, \quad (7)$$

where $f(E(t))$ is the cost of running the campaign at intensity $E(t)$, and seek $E^* \in \Omega$ such that $J(E^*) = \min_{E \in \Omega} J(E)$, with control set

$$\Omega = \{E(t) \in L^1(0, t_f) : 0 \leq E(t) \leq E_{\max}\}.$$

The Hamiltonian and adjoint equations

By Pontryagin's Maximum Principle, we construct the Hamiltonian

$$\begin{aligned} H = & M + f(E) + \lambda_1(-\alpha_1 ES + \psi C - \beta SM + \theta M) \\ & + \lambda_2(\alpha_1 ES - \psi C + \gamma CM + \alpha_2 EM) \\ & + \lambda_3(\beta SM - \gamma CM - \theta M - \alpha_2 EM - dM), \end{aligned} \quad (8)$$

where $\lambda_1, \lambda_2, \lambda_3$ are the adjoint variables. Differentiating H with respect to the state variables gives the adjoint system, which is independent of the choice of cost function f :

$$\frac{d\lambda_1}{dt} = -\frac{\partial H}{\partial S} = \lambda_1(\alpha_1 E + \beta M) - \lambda_2(\alpha_1 E) - \lambda_3(\beta M), \quad (9)$$

$$\frac{d\lambda_2}{dt} = -\frac{\partial H}{\partial C} = -\lambda_1\psi + \lambda_2(\psi - \gamma M) + \lambda_3(\gamma M), \quad (10)$$

$$\frac{d\lambda_3}{dt} = -\frac{\partial H}{\partial M} = -1 + \lambda_1(\beta S - \theta) - \lambda_2(\gamma C + \alpha_2 E) - \lambda_3(\beta S - \gamma C - \theta - \alpha_2 E - d), \quad (11)$$

with transversality conditions $\lambda_1(t_f) = \lambda_2(t_f) = \lambda_3(t_f) = 0$.

Optimal control characterization

Isolating the terms of H containing E and defining the marginal-benefit function

$$\Phi(t) = \alpha_1 S(\lambda_1 - \lambda_2) + \alpha_2 M(\lambda_3 - \lambda_2), \quad (12)$$

which represents the instantaneous value of shifting individuals into the critically-literate compartment, the partial derivative of H with respect to the control is

$$\frac{\partial H}{\partial E} = f'(E) - \Phi(t). \quad (13)$$

We consider two cost structures.

Case 1: Linear cost (bang-bang control). If $f(E) = W_1 E$ for a constant weight W_1 , then $\partial H / \partial E = W_1 - \Phi(t)$, and since H is linear in E , the optimal control takes bang-bang form:

$$E^*(t) = \begin{cases} E_{\max} & \text{if } \Phi(t) > W_1, \\ 0 & \text{if } \Phi(t) < W_1, \\ E_{\text{singular}} & \text{if } \Phi(t) = W_1 \text{ on a finite interval.} \end{cases} \quad (14)$$

Biologically, this says a policymaker should apply maximum educational effort whenever its marginal benefit exceeds its fixed per-unit cost W_1 , and none otherwise.

Remark 3 (Absence of a singular arc). The characterization (14) leaves open whether the singular case $\Phi(t) = W_1$ occurs on a nontrivial interval. Following Joshi et al. (2015), we differentiate the switching function $\partial H / \partial E = W_1 - \Phi(t)$ along the state–adjoint trajectory. A direct computation shows the control E cancels identically from the first derivative, $\frac{d}{dt}(\partial H / \partial E)$, so a putative singular arc is of order one; the control reappears only in the second derivative,

$$\frac{d^2}{dt^2} \left(\frac{\partial H}{\partial E} \right) = \Phi_1(t) E(t) + \Phi_2(t), \quad E_{\text{singular}}(t) = -\frac{\Phi_2(t)}{\Phi_1(t)}, \quad (15)$$

which would be the singular control were an arc to occur, admissible only where it lies in $[0, E_{\max}]$ and satisfies the generalized Legendre–Clebsch condition $\Phi_1(t) < 0$. For the calibrated parameters, however, no singular arc arises: evaluated along the optimal trajectory, the switching function $\Phi(t) - W_1$ has only isolated, transversal zeros (its time-derivative is nonzero at each crossing, and since E has already dropped out of that derivative, the crossing is well-defined independent of the control), so the trajectory passes through $\Phi = W_1$ instantaneously rather than remaining on it. The optimal linear-cost control is therefore purely bang-bang, with a single front-loaded switch. This also explains the numerical chattering seen with a naive discontinuous sweep as an artifact of the update scheme rather than a genuine singular arc.

Case 2: Quadratic cost (continuous control). If $f(E) = \frac{1}{2} W_2 E^2$, representing escalating marginal costs as the campaign scales up, then $\partial H / \partial E = W_2 E - \Phi(t)$, giving the continuous characterization

$$E^*(t) = \min \left\{ E_{\max}, \max \left\{ 0, \frac{\alpha_1 S^*(\lambda_1 - \lambda_2) + \alpha_2 M^*(\lambda_3 - \lambda_2)}{W_2} \right\} \right\}. \quad (16)$$

Unlike the bang-bang case, this characterization yields a strategy that smoothly scales up during peak virality and tapers off as the threat decays.

Existence of an optimal control

Theorem 1 (Existence). *For each cost structure above (linear or quadratic), there exists an optimal control $E^*(t) \in \Omega$ and corresponding state trajectory (S^*, C^*, M^*) minimizing $J(E)$ in (7).*

Proof sketch. We verify the standard conditions of the Fleming–Rishel existence theorem (Fleming and Rishel, 1975), as applied to closely analogous SIR-type educational-campaign models by Joshi et al. (2015) and to noncompliance-type SIR models by Ngo et al. (2025).

- (i) *Nonempty admissible set.* By Proposition 2, for any measurable $E(t) \in \Omega$ the corresponding state trajectory exists, is unique (the right-hand side of (1)–(3) is continuous and Lipschitz in the states, uniformly in $E \in [0, E_{\max}]$), and remains in the compact region Γ , so the set of admissible state-control pairs is nonempty.

- (ii) *Convex, closed control set.* $\Omega = \{E : 0 \leq E(t) \leq E_{\max}\}$ is convex, closed, and bounded.
- (iii) *Linear state dynamics in the control.* The right-hand side of (1)–(3) is affine in E for fixed states, and is bounded by a linear function of the state and control variables on $\Gamma \times \Omega$.
- (iv) *Convexity of the integrand.* The integrand $M(t) + f(E(t))$ of J is convex in E on Ω : in the linear case $f(E) = W_1 E$ is (trivially) convex and Ω is compact, so existence follows directly from compactness and continuity (the Filippov–Cesari theorem) without a growth condition; in the quadratic case $f(E) = \frac{1}{2} W_2 E^2$ is strictly convex and satisfies the coercivity bound $f(E) \geq c_1 |E|^2 - c_2$ for constants $c_1, c_2 > 0$, satisfying the growth condition of Fleming and Rishel (1975).

Conditions (i)–(iv) satisfy the hypotheses of Fleming and Rishel (1975, Theorem III.4.1) (see also Lukes, 1982, Corollary 4.1, for the compact-control-set case), establishing existence of a minimizer. $\square$

Remark 4 (Uniqueness). As in Joshi et al. (2015), uniqueness of the optimal control can be established for t_f sufficiently small, via a Grönwall-type bound on the difference between two candidate optimal state–adjoint trajectories. In the numerical simulations we observe no indication of non-uniqueness, and no singular arc occurs (Remark 3).

References

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